

MAL Omics: Proteomics Analysis Report

COPDGene, SomaScan (~4,979 proteins), $n = 1,306$

Study. COPDGene is a longitudinal cohort study of current and former smokers enrolled at 21 US clinical centers. Subjects were seen at Phase 2 (~2013) and Phase 3 (~2019). Blood was drawn at Phase 2 and profiled by SomaScan, an aptamer-based platform that quantifies ~4,979 plasma proteins simultaneously as RFU (Relative Fluorescence Units). This analysis uses $n = 1,306$ subjects with complete proteomics, phenotype, and MAL data.

MAL (Mechanically Affected Lung). Computed tomography (CT)-derived continuous score (range 0.01–0.40) measuring the fraction of lung volume under destructive mechanical stress. Higher MAL = more lung tissue under stress. MAL is computed from inspiratory/expiratory CT image registration and is distinct from standard emphysema measures (e.g., %LAA-950). Variable name in data: `meanmal`.

MAL is itself a strong predictor of who will develop worse emphysema over time: higher mechanical stress on lung tissue drives progressive structural damage. However, MAL requires specialized CT analysis and is not available from a routine blood draw.

Goal. (1) Identify which Phase 2 plasma proteins are associated with MAL, establishing that mechanical lung stress leaves a detectable signal in blood. (2) Use those proteins to build a Proteomic Risk Score (PRS) from a Phase 2 blood draw that approximates MAL and can predict who will develop worse emphysema by Phase 3. If successful, the PRS would allow MAL-like risk stratification without requiring CT image registration. The proteins themselves are not measured at Phase 3; they serve as baseline predictors of the CT density change that occurs between visits (~6 years apart).

Outcome. Emphysema progression = change in CT lung density Phase 3 – Phase 2 (`Change_lung_density_vnb_P2_P3`, Hounsfield Units [HU]). Negative = worsened. Mean ≈ 1 HU (standard deviation [SD] 7) in this cohort; a negative value means the lung became less dense (more emphysematous) between visits.

Data Sources and Variable Names

Four CSV files were used. All joins are on `sid`.

`data/csv_exports/prot.csv`: SomaScan proteomics

Column	Description
<code>sid</code>	Subject identifier (join key)
<code>X{seqID}</code>	Raw RFU value for each protein (e.g. <code>X10000.28</code>). ~4,979 columns matching pattern <code>^X[0-9]</code> . Renamed to <code>log2_X{seqID}</code> after $\log_2$ transform.

data/csv_exports/pheno_MAL.csv: **Phenotypes and covariates**

Column	Used as
sid	Join key
Change_lung_density_vnb_P2_P3	Outcome: emphysema progression (HU)
lung_density_vnb_P2	Baseline lung density (HU)
Age_P2	Age (years)
gender	Sex (coded 1/2)
race	Race (coded 1/2)
ATS_PackYears_P2	Pack-years of smoking
SmokCigNow_P2	Current smoking (0/1)
scanner_model_clean_P2	CT scanner model
FEV1_FVC_post_P2	FEV ₁ (forced expiratory volume in 1 second) / FVC (forced vital capacity)
PC1 – PC5	Ancestry principal components
FEV1pp_P2	FEV ₁ % predicted
BMI / bmi_P2	Body mass index

data/Data/MAL.csv: **MAL scores**

Column	Description
sid	Subject identifier (join key)
meanmal	MAL score (continuous, range 0.01–0.40)

data/csv_exports/metadat5k.csv: **Protein annotation**

Column	Description
seqID	Protein sequence identifier; joins to $\log_2 X_{\{\text{seqID}\}}$
EntrezGeneSymbol	HGNC (HUGO Gene Nomenclature Committee) gene symbol (e.g. ADIPOQ)
Target	Short protein name (e.g. Adiponectin)
TargetFullName	Full protein name

Join logic. Three-way inner join: `prot` $\bowtie_{\text{sid}}$ `pheno` $\bowtie_{\text{sid}}$ `MAL`. Metadata joined on `seqID` = `sub("^log2.X", "", predictor)`.

File sizes: `prot.csv` has 5,670 subjects; `MAL.csv` has 5,024 subjects; `pheno_MAL.csv` has 10,652 subjects. After the 3-way inner join on `sid`: $n = 4,630$.

Complete-case filter then requires non-missing values for all of: `Change_lung_density_vnb_P2_P3` (Phase 3 outcome), `lung_density_vnb_P2`, `meanmal`, `Age_P2`, `gender`, `race`, `ATS_PackYears_P2`, `SmokCigNow_P2`, `scanner_model_clean_P2`, `FEV1_FVC_post_P2`, and `PC1–PC5`.

This reduces the sample to $n = 1,306$. The large drop ($4,630 \rightarrow 1,306$) is driven primarily by the Phase 3 outcome: 8,912 of 10,652 subjects in `pheno_MAL.csv` (84%) have no Phase 3 CT scan recorded, meaning they did not return for the follow-up visit ~ 6 years later.

Table 1: Demographics

Study population after 3-way merge (proteomics $\times$ phenotype $\times$ MAL) and complete-case filtering.

Variables not found in the phenotype file:

- *Not found: FEV1pp_P2 (FEV₁% predicted)*: column not present under any searched pattern (FEV1pp, FEV1pct, FEV1_percent_pred). Row left blank.
- *Not found: BMI*: no column matching $\sim$ [Bb] [Mm] [Ii] found in the phenotype file. Row left blank.

Table 1. Baseline characteristics of COPDGene participants included in the proteomics analysis ($n = 1,306$). Subjects had complete SomaScan plasma proteomics, phenotype data, and MAL score. Continuous variables: mean (SD); categorical: n (%). MAL (Mechanically Affected Lung) = CT-derived score (range 0.01–0.40); higher values indicate more lung under destructive mechanical stress. Emphysema progression = CT lung density at Phase 3 minus Phase 2 ($\rho_{P3} - \rho_{P2}$, HU); negative values indicate worsening. FEV₁% predicted and BMI were not available in the phenotype file.

Characteristic	Overall ($n = 1,306$)
<i>Demographics</i>	
Age (years)	65 (8)
Sex: male / female	633 (48%) / 673 (52%)
Race: White / Black	991 (76%) / 315 (24%)
<i>Smoking</i>	
Current smoker	445 (34%)
Pack-years	43 (23)
<i>Pulmonary function</i>	
FEV ₁ % predicted	<i>Not found: column not found in phenotype file</i>
FEV ₁ /FVC	0.69 (0.14)
<i>CT imaging</i>	
Lung density Phase 2 (HU)	83 (23)
MAL score	0.04 (0.05)
Emphysema progression (HU)	1 (7)
<i>Anthropometrics</i>	
BMI (kg/m ²)	<i>Not found: column not found in phenotype file</i>

Note. Sex is coded 1/2 in the data; assumed Male/Female per COPDGene convention. Race coded 1/2; assumed White/Black. Emphysema progression mean of 1 HU (SD 7) reflects a near-zero mean; the distribution includes both progressors (negative $\Delta\rho$) and improvers (positive $\Delta\rho$).

Figure 1: Study Design

Not automatically generated. Requires a manual schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics $\rightarrow$ differential expression $\rightarrow$ GSEA (Gene Set Enrichment Analysis) / STRING $\rightarrow$ adaptive LASSO (Least Absolute Shrinkage and Selection Operator) PRS (Proteomic

Risk Score) → emphysema progression association → causal mediation.

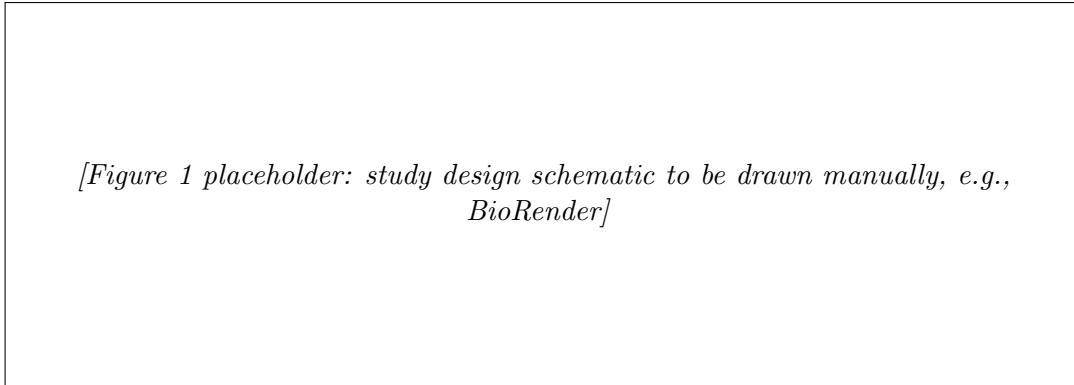


Figure 1: Overview of the COPDGene MAL omics analysis pipeline. SomaScan plasma proteomics ($\sim 4,979$ proteins, Phase 2) are used to identify proteins associated with MAL (differential expression, GSEA, STRING network), build a proteomic risk score (adaptive LASSO), test whether the score predicts emphysema progression (Phase 3 minus Phase 2 CT density), and quantify mediation of the MAL effect through blood proteins (causal mediation analysis). Schematic to be drawn manually.

Figure 2: Protein Expression Landscape of MAL

A. Volcano Plot (protein $\sim$ MAL)

Each dot is one protein. The x -axis ($\hat{\beta}$) shows how much a protein changes per one-unit increase in MAL after adjusting for age, sex, race, smoking, pack-years, scanner, and ancestry PCs. Positive $\hat{\beta}$ = protein is higher with more mechanical lung stress; negative = lower. $\hat{\beta}$ is defined as the change in $\log_2$ protein per *one-unit* increase in MAL. But MAL only ranges from 0.01 to 0.40 in this cohort, so a one-unit change never actually occurs. To express the effect in realistic terms: multiplying $\hat{\beta}$ by the full MAL range ($0.40 - 0.01 = 0.39$) gives the maximum expected protein change across the entire observed spread of MAL. For Adiponectin: $0.27 \times 0.39 \approx 0.1 \log_2$ -unit shift from the lowest to the highest MAL subject. The y -axis ($-\log_{10}$ FDR [false discovery rate]) is confidence: higher = more significant. Dots above the threshold ($\text{FDR} < 0.05$) are coloured red (enriched) or blue (depleted); grey dots did not reach significance. A dot far right and high = reliably elevated with more MAL. A dot far left and high = reliably lower with more MAL.

How calculated. For each of 4,979 proteins:

$$\log_2(\text{protein}) \sim \beta_{\text{MAL}} \cdot \text{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1-PC5}$$

$\hat{\beta}_{\text{MAL}}$ = change in $\log_2$ protein level per one-unit increase in MAL. FDR correction (Benjamini-Hochberg) across all 4,979 tests.

Key result. 45 of 4,979 proteins significant at $\text{FDR} < 0.05$.

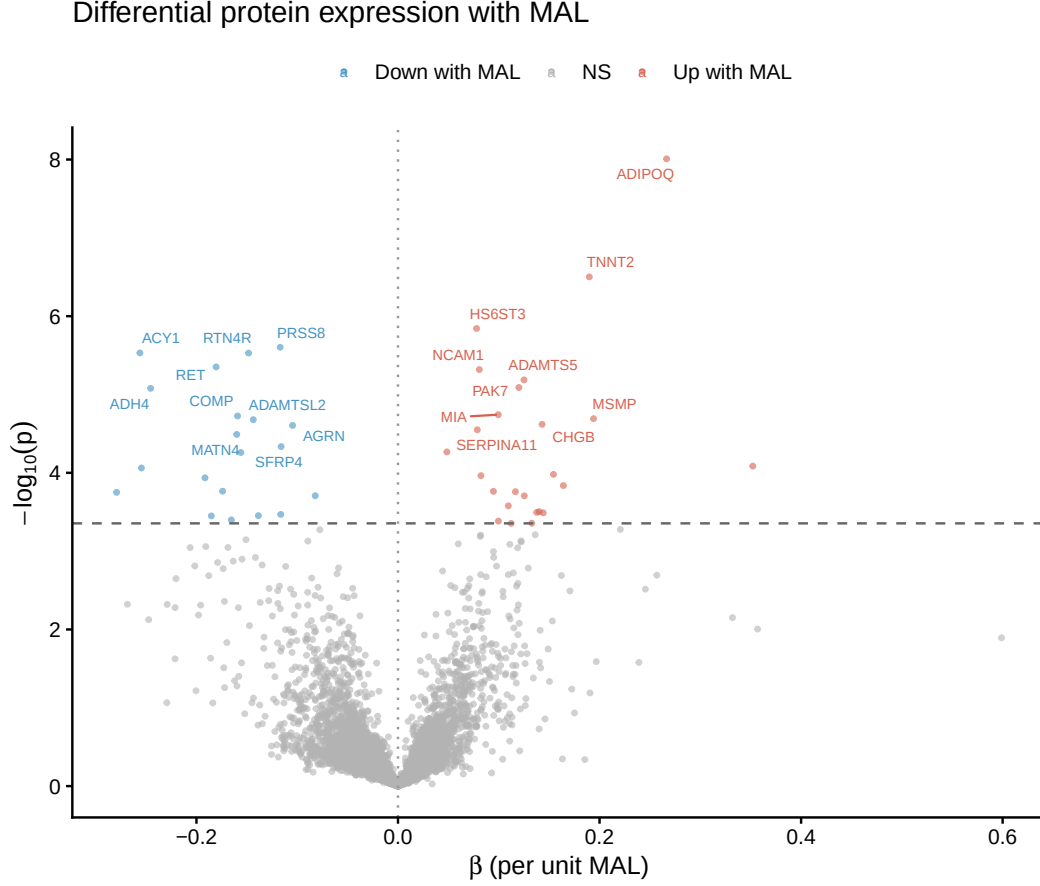


Figure 2: Differential plasma protein expression associated with Mechanically Affected Lung (MAL) score in COPDGene participants ($n = 1,306$; 4,979 SomaScan proteins). Each point is one protein. x -axis: regression coefficient ($\hat{\beta}$) for $\log_2$ protein level per unit MAL, adjusted for age, sex, race, pack-years, smoking, CT scanner, and ancestry PCs (PC1–PC5). y -axis: $-\log_{10}(\text{FDR}; \text{Benjamini-Hochberg})$. Red = $\text{FDR} < 0.05$, positively associated; blue = $\text{FDR} < 0.05$, negatively associated; grey = not significant. Top 20 proteins by FDR labelled. 45 proteins reached $\text{FDR} < 0.05$.

Top enriched (higher with more MAL): *ADIPOQ* (Adiponectin, $\beta = 0.27$), *TNNT2* (Troponin T, $\beta = 0.19$), *ADAMTS5*, *NCAM1*, *PAK7*.

Top depleted (lower with more MAL): *ACY1* ($\beta = -0.26$), *ADH4* ($\beta = -0.25$), *RTN4R* (Nogo Receptor), *RET*, *PRSS8*, *SFRP4*.

B. GSEA Barplot

GSEA asks whether an entire biological pathway shows a coordinated shift with MAL, rather than asking protein by protein. Each bar is one significant pathway. Bar pointing right ($\text{NES} > 0$): the pathway's proteins are on average higher in plasma with higher MAL. Bar pointing left ($\text{NES} < 0$): lower. Longer bar = stronger shift. NES of ± 2 is a strong, consistent signal. padj is the Benjamini-Hochberg adjusted p -value across all pathways tested; only bars with $\text{padj} < 0.05$ are shown.

How calculated. All 4,979 proteins ranked by their MAL t -statistic (most positively associated at

top, most negatively at bottom). Gene sets tested: Hallmark, KEGG Legacy, Reactome (gene-set sizes 15–500).

NES (Normalized Enrichment Score): measures whether a gene set’s members concentrate at the top or bottom of the ranked list. $NES > 0$ = enriched in high-MAL subjects; $NES < 0$ = depleted. Normalization accounts for gene-set size so scores are comparable across pathways.

padj (adjusted p -value): fgsea tests hundreds of pathways at once. Without correction, many would appear significant by chance. padj is the Benjamini–Hochberg correction applied within each gene set collection (Hallmark, KEGG, Reactome separately). A $padj < 0.05$ means fewer than 5% of pathways at this threshold are expected to be false positives. Only pathways with $padj < 0.05$ are shown in the barplot.

Results. 9 pathways significant ($padj < 0.05$). Seven are depleted ($NES < 0$): oxidative phosphorylation, glycolysis, fatty acid metabolism, amino acid handling, and xenobiotic processing. Two are enriched ($NES > 0$): chemokine receptor signalling and antimicrobial peptides (both inflammatory/immune). Higher MAL is associated with lower circulating metabolic proteins and higher inflammatory proteins.

Gene Set Enrichment Analysis (GSEA): significant pathways associated with MAL. Proteins ranked by MAL t -statistic ($n = 4,979$) and tested using fgsea against Hallmark, KEGG Legacy, and Reactome gene sets (sizes 15–500). NES = Normalized Enrichment Score; $NES > 0$ = pathway proteins higher in high-MAL subjects (enriched); $NES < 0$ = lower (depleted). padj = Benjamini–Hochberg adjusted p -value within each collection. Only pathways with $padj < 0.05$ shown ($n = 9$).

Collection	Pathway	NES	padj
Hallmark	Fatty Acid Metabolism	−1.76	0.021
Hallmark	Oxidative Phosphorylation	−1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	−2.02	0.018
Reactome	Biological Oxidations	−2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	−1.84	0.013
Reactome	Diseases of Metabolism	−1.76	0.032
Reactome	Phase I Functionalization of Compounds	−1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

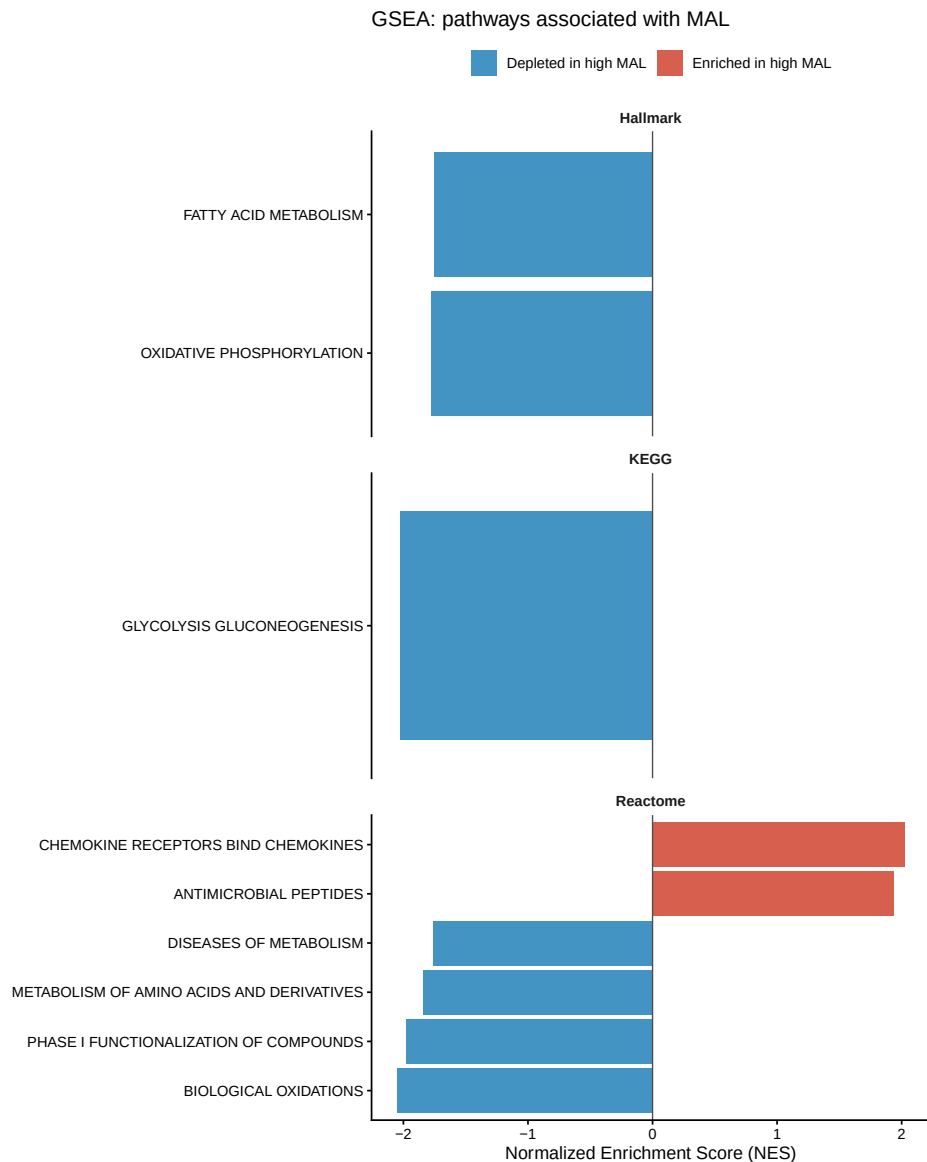


Figure 3: Gene Set Enrichment Analysis (GSEA) of MAL-associated plasma proteins across Hallmark, KEGG Legacy, and Reactome collections. Proteins ranked by MAL t -statistic ($n = 4,979$); bar length represents the Normalized Enrichment Score (NES). NES > 0 = enriched in high-MAL subjects; NES < 0 = depleted. Colour indicates direction. Only pathways with Benjamini–Hochberg adjusted $p < 0.05$ are shown ($n = 9$; 7 depleted, 2 enriched).

C. STRING Network

Protein–protein interaction network for the 45 MAL-associated proteins. Each node is one protein. An edge between two proteins means they are known to physically interact or co-function in humans per STRING. A cluster of connected nodes indicates those proteins share a biological process. A hub (many edges) may be a central regulator in the MAL response. An isolated node had no high-confidence interactions with the other significant proteins. Edge confidence ≥ 400 (medium confidence).

How calculated. 45 FDR-significant proteins submitted to STRING v11.5 human database. Interactions with confidence ≥ 400 drawn. Only PNG available: `STRINGdb::plot_network()` does not support PDF output.

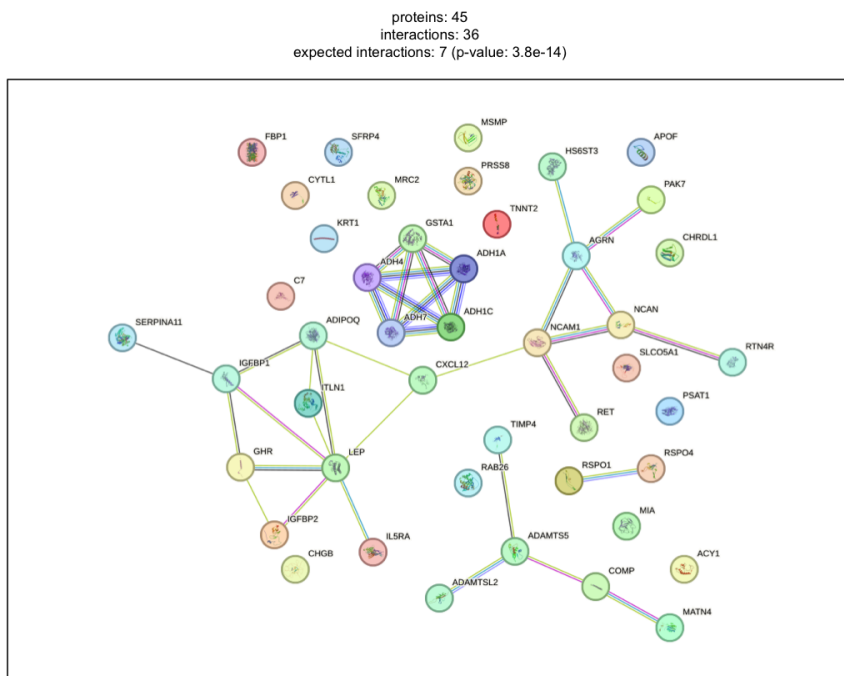


Figure 4: Protein–protein interaction network for the 45 plasma proteins significantly associated with MAL (FDR < 0.05), constructed using STRING v11.5 (human, species 9606). Nodes represent proteins; edges indicate known or predicted functional interactions with confidence score ≥ 400 (medium confidence). Isolated nodes had no qualifying interactions within this set. Network image exported as PNG; PDF output is not supported by `STRINGdb::plot_network()`.

Figure 3: Quartile Plot of Proteomic Risk Score

Subjects divided into four groups by Proteomic Risk Score. Quartile 1 = lowest 25% PRS (blood protein pattern least like high MAL). Quartile 4 = highest 25% (most like high MAL). The y -axis ($\hat{\beta}$, HU/SD) is the OLS (ordinary least squares) coefficient for PRS in that quartile subgroup: positive = higher PRS associated with less density loss; negative = more density loss; zero = no association. Error bars are 95% CI (confidence interval); bars crossing the dashed zero line are not statistically significant.

Key result. Effect sizes are small and confidence intervals cross zero in all quartiles. The PRS does not predict emphysema progression within any risk stratum ($p = 0.079$ overall).

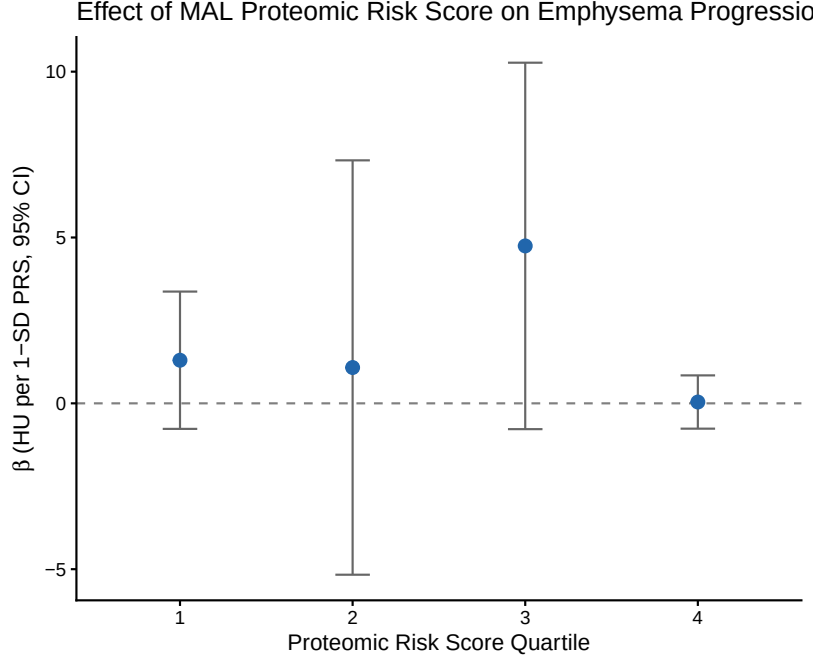


Figure 5: Association between the Proteomic Risk Score (PRS) and emphysema progression stratified by PRS quartile ($n = 1,306$). Subjects were divided into quartiles by PRS (Q1 = lowest 25%, Q4 = highest 25%). Within each quartile, OLS regression estimated the $\hat{\beta}$ (HU per SD of PRS) for the PRS–progression association, adjusted for age, sex, race, pack-years, smoking, and CT scanner. Points show $\hat{\beta}$ with 95% confidence intervals. Dashed line at zero = no association. Overall PRS $p = 0.079$.

Proteomic Risk Score: Methods and Results

The PRS is a weighted sum of 378 plasma proteins predicting MAL. High PRS = blood protein pattern similar to high MAL (more mechanically stressed lung). Low PRS = pattern similar to low MAL. Test $R^2 = 0.157$: the PRS explains 15.7% of MAL variation in held-out subjects; 84% is not captured by the blood proteome. Positive $\hat{\beta}$ in the weights = protein higher in people with more MAL. Negative $\hat{\beta}$ = protein lower with more MAL.

Data split. 70/30 train/test (`set.seed(2024)`), $n_{\text{train}} = 914$, $n_{\text{test}} = 392$.

Step 1: Log2 transform all protein RFU values

$$\tilde{x} = \log_2(\max(x, 10^{-6}))$$

Transforms skewed RFU values to approximately normal scale. The 10^{-6} floor prevents $\log(0)$ errors.

Step 2: Adaptive weights from ridge regression

$$w_j = \frac{1}{|\hat{\beta}_j^{\text{ridge}}|}$$

Ridge regression is fit on the residualised MAL outcome (after projecting out the unpenalised covariates). Each protein j gets a weight inversely proportional to its ridge coefficient. Proteins

that ridge identified as informative get lower penalty in the LASSO step; irrelevant proteins get higher penalty. This gives the adaptive LASSO its oracle property (tendency to select the true model).

Step 3: Adaptive LASSO selection

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \sum_j w_j |\beta_j|$$

λ chosen by 10-fold cross-validation ($\lambda_{\min}$).

Covariates (age, sex, race, smoking, pack-years, scanner, PC1–PC5) are assigned penalty factor $w_j = 0$. Here w_j is not the covariate’s coefficient; it is the multiplier on the L1 penalty term for that variable. When $w_j = 0$, the penalty contribution for that variable is $\lambda \times 0 \times |\beta_j| = 0$, meaning the LASSO applies no shrinkage to it. The covariate is estimated freely (like ordinary least squares) and is always retained in the model regardless of λ . Proteins, by contrast, have $w_j > 0$ (from the ridge pre-step), so their coefficients are penalised and can be shrunk to zero and dropped. This ensures confounders are always adjusted for and only proteins compete for selection.

Step 4: Post-selection OLS refit. Selected proteins refit by ordinary least squares. LASSO shrinks coefficients toward zero; OLS on the selected set gives unbiased estimates, proper SE (standard error), CI, and p -values.

Step 5: Compute the risk score for each subject

$$\text{PRS}_i = \sum_{j \in \mathcal{S}} \hat{\beta}_j^{\text{OLS}} \cdot \tilde{x}_{ij}$$

where $\mathcal{S}$ is the selected protein set. Standardised to mean 0, SD 1 across the full cohort so the regression coefficient has units of HU per SD of the score.

Result. 378 proteins selected. Test $R^2 = 0.157$.

Top 20 proteins in the Proteomic Risk Score (PRS) by absolute OLS coefficient. The PRS was derived by adaptive LASSO (ridge pre-step for penalty weights, 10-fold cross-validation at $\lambda_{\min}$) predicting MAL score in the training set ($n = 914$, 70%), followed by post-selection OLS refit. Of 4,979 proteins, 378 were selected (test $R^2 = 0.157$). $\hat{\beta}$ = OLS coefficient ($\log_2$ RFU change per unit MAL); 95% CI and p -values from post-selection OLS.

Gene	Target	$\hat{\beta}$	95% CI (low)	95% CI (high)	p
ZBPB	ZBPB1	+0.286	0.160	0.412	< 0.001
CFI	Factor I	-0.226	-0.398	-0.054	0.010
GREM2	GREM2	+0.193	0.113	0.273	< 0.001
C7	Complement C7	+0.169	0.030	0.309	0.017
SEC13	SEC13	-0.153	-0.254	-0.053	0.003
SERPINF2	α_2 -Antiplasmin	+0.149	0.011	0.287	0.035
FSTL1	FSTL1	-0.147	-0.309	0.015	0.075
APOH	β_2 -Glycoprotein I	+0.143	-0.049	0.335	0.145
COL3A1	Collagen Type III	+0.132	0.061	0.204	< 0.001
C3	C3a	+0.131	0.021	0.242	0.020
SMS	Spermine Synthase	-0.125	-0.179	-0.072	< 0.001
MATN2	MATN2	-0.125	-0.242	-0.008	0.036
GALNT11	GLT11	-0.122	-0.225	-0.019	0.021
ROBO2	ROBO2	-0.119	-0.220	-0.017	0.022
SEZ6L	SEZ6L	+0.117	0.020	0.214	0.019
FBLN1	Fibulin-1	-0.113	-0.278	0.051	0.175
ADAMTS5	ADAMTS-5	+0.113	0.062	0.163	< 0.001
CCL23	MPIF-1	-0.112	-0.160	-0.063	< 0.001
CREB3L4	CR3L4	+0.108	0.047	0.169	< 0.001
CNTN1	Contactin-1	-0.107	-0.229	0.015	0.086

AIC Comparison: PRS vs. Clinical Model

Three models predicting emphysema progression are compared by AIC (Akaike Information Criterion: $-2\log L + 2k$). Lower AIC = better model. The clinical model is the reference; Δ AIC for the other models is their AIC minus the clinical AIC. A Δ AIC of ≤ -2 means the new model is meaningfully better than the clinical reference; a positive Δ AIC means it is worse.

- **Clinical:** age, sex, race, smoking status, pack-years, CT scanner type.
- **PRS only:** protein score, baseline lung density, scanner.
- **Clinical + PRS:** clinical variables plus the protein score.

PRS $\hat{\beta} = 0.42$ HU/SD ($p = 0.079$) in the combined model: for each 1 SD increase in PRS, emphysema progression changes by 0.42 HU on average. Given a progression SD of ~ 7 HU, this is a small, non-significant effect.

Note. BMI was not available in the phenotype file and is absent from the clinical model. AIC comparisons may change once BMI is added.

Model comparison for predicting emphysema progression by AIC (Akaike Information Criterion; lower = better fit). Three OLS models predicting Δ CT lung density (HU) were compared. Δ AIC is relative to the clinical reference model; a difference of ≤ -2 indicates meaningful improvement over the clinical reference. Δ AIC = model AIC minus clinical AIC; positive values indicate a worse fit than clinical alone. PRS = 378-protein adaptive LASSO risk score, standardised to mean 0, SD 1. BMI was unavailable and is excluded from the clinical model.

Model	AIC	Δ AIC
Clinical (age, sex, race, smoking, pack-years, scanner)	8,815	Reference
PRS only (PRS + baseline density + scanner)	8,848	+33.1
Clinical + PRS	8,816	+0.8

PRS $\hat{\beta} = 0.42$ HU/SD, 95% CI $[-0.05, 0.88]$, $p = 0.079$ in the combined model.

Key result. The PRS does not improve model fit over clinical variables alone (Δ AIC = +0.8, far from the -2 threshold). The PRS-only model is substantially worse than clinical alone (Δ AIC = +33).

Table 2: Causal Mediation

The analysis decomposes the total effect of MAL on emphysema progression into a direct path (NDE) and a path through the blood protein score (NIE). NDE = how much of the MAL effect goes through pathways other than the protein score. NIE = how much goes through the protein score. All effects are in HU (Hounsfield Units), the CT density scale. More negative HU = lower lung density = more emphysema. A difference of 1 HU is a small change; the SD of progression in this cohort is ~ 7 HU.

- **NDE = 19.7 HU**, $p = 0.004$: MAL drives progression through mechanisms independent of the protein score. A one-unit increase in MAL is associated with 19.7 HU more progression via the direct path.
- **NIE = -0.12 HU**, $p = 0.97$: essentially zero and non-significant. The proteins are not the mechanism by which MAL causes progression.
- **Proportion mediated = -0.6%** : of the total MAL effect, -0.6% is explained by the blood protein pathway. Negative proportion arises when the indirect effect goes in the opposite direction to the total effect; here it reflects noise around zero.
- **Total effect = 19.6 HU**: NDE + NIE, almost entirely direct.

How calculated. Pathway tested: MAL $\rightarrow$ Proteomic Risk Score $\rightarrow$ Emphysema Progression. Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

Two models fitted internally by `medflex`:

Imputation model: $\Delta\rho \sim \text{meanmal} \times \text{PRS} + \text{covariates}$

Natural effects model: $\Delta\rho \sim \text{meanmal}_0 + \text{meanmal}_1 + \text{covariates}$

meanmal_0 is the MAL value at which the mediator (PRS) is set (direct path); meanmal_1 is the

actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Causal mediation of the MAL–emphysema progression relationship through the Proteomic Risk Score (PRS), estimated using the natural effects framework (`medflex` R package, robust sandwich SE, $n = 1,306$). NDE (Natural Direct Effect) = portion of MAL’s effect on Δ CT lung density not operating through the PRS. NIE (Natural Indirect Effect) = portion operating through the PRS. Proportion mediated = NIE / total effect. All effects in HU (Hounsfield Units; negative = more emphysema).

Effect	Estimate (HU)	z	p
Natural Direct Effect (NDE)	+19.72	2.92	0.004
Natural Indirect Effect (NIE)	−0.12	−0.04	0.971
Total effect	+19.60	N/A	N/A
Proportion mediated	−0.6%		

Key result. MAL has a large, significant direct effect on emphysema progression (NDE $p = 0.004$). The proteomic risk score mediates essentially none of this effect (NIE $p = 0.97$, proportion = −0.6%).

Supplement

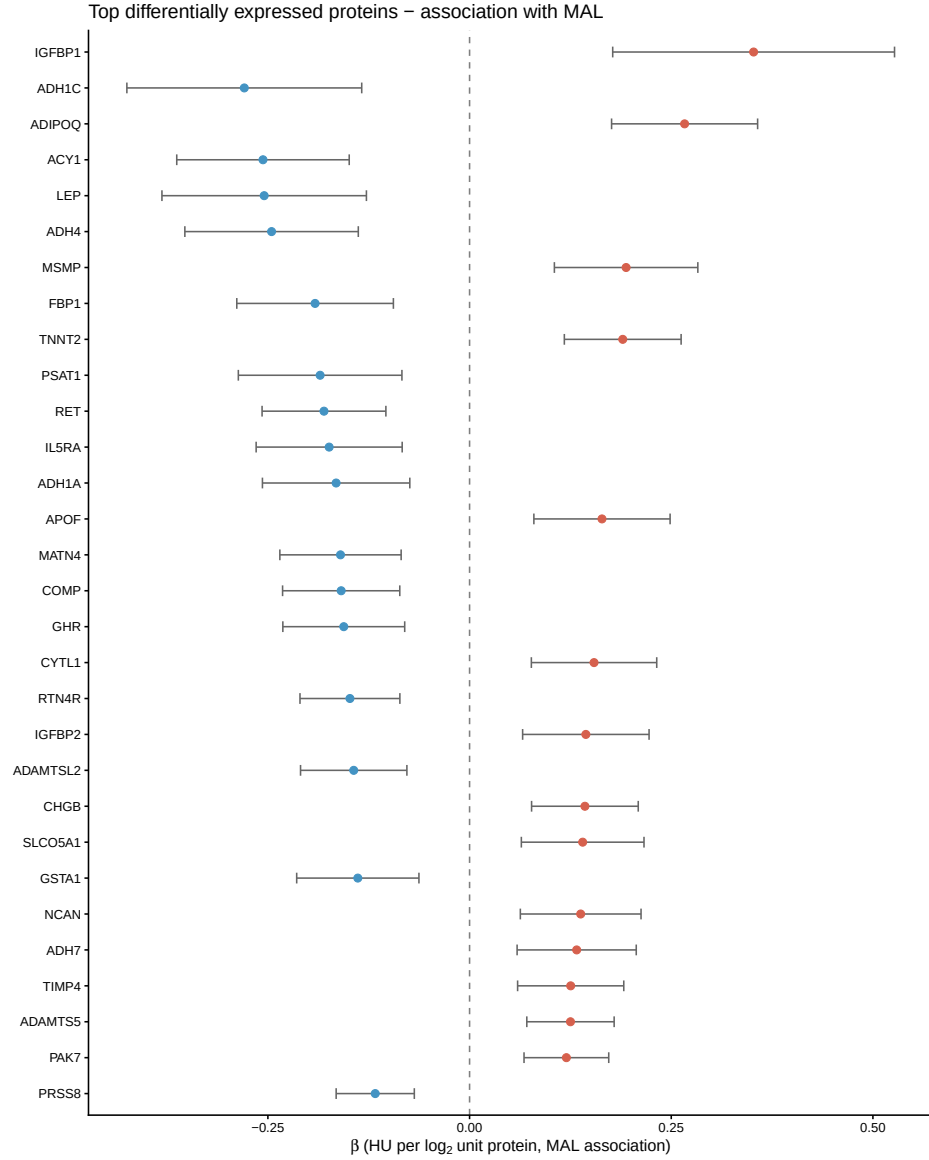
Supplementary Tables

- **S1: All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t , p , FDR.
File: MAL_omics/mal_de_results.csv
- **S2: PRS weights** (378 proteins): gene, target, $\hat{\beta}_{OLS}$, 95% CI, p .
File: MAL_omics/mal_risk_score_weights.csv
- **S3: Per-protein mediation** (378 proteins): NDE, NIE, total, proportion mediated, NIE p .
File: MAL_omics/mal_mediation_supp.csv

Top 6 individual mediators (NIE $p < 0.05$): TREM2 (13%, $p = 0.017$), ADAMTS5 (5.7%, $p = 0.021$), SEZ6L (17%, $p = 0.025$), RSPO1 (14%, $p = 0.038$), GREM2 (12%, $p = 0.039$), ADH7 (7%, $p = 0.044$). All small in absolute magnitude.

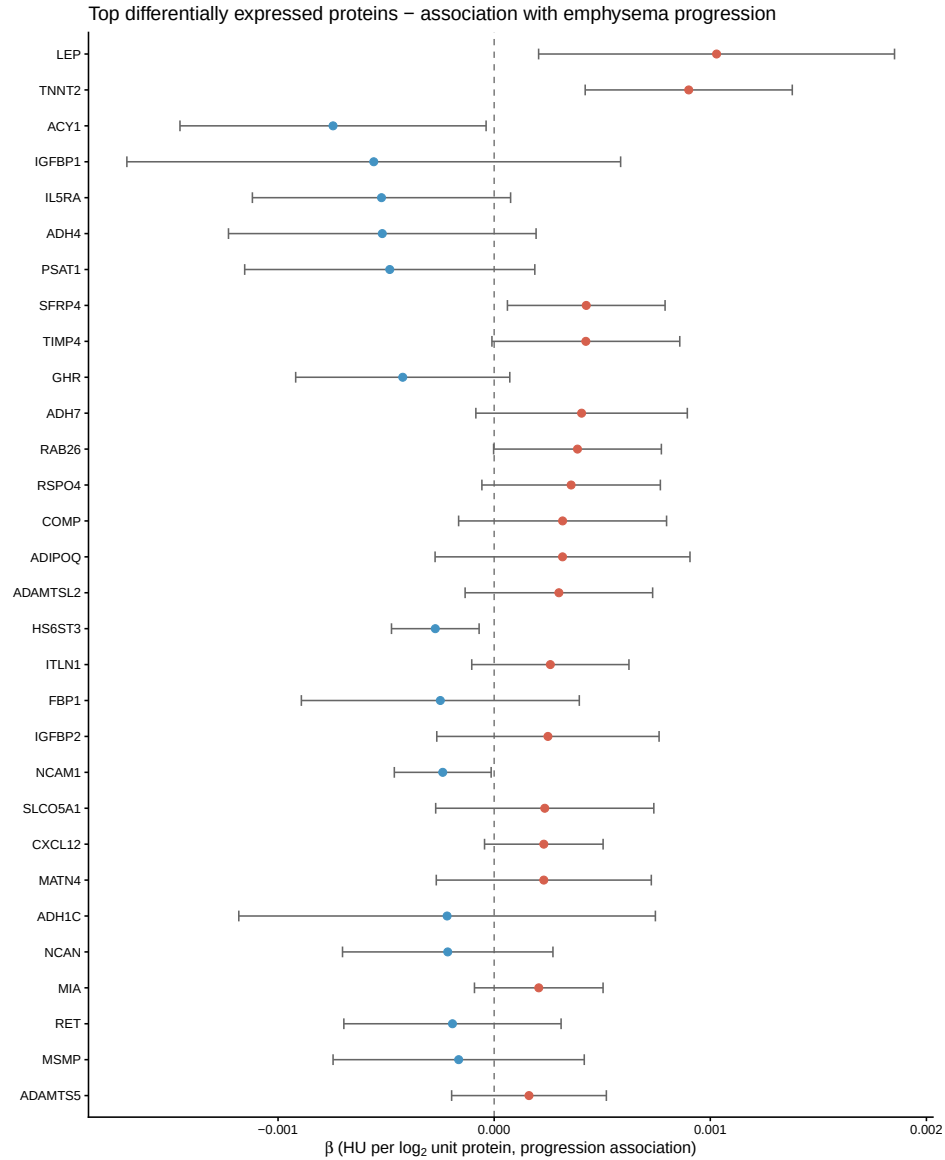
Supplementary Figures

- **Supp Figure A: Forest plot, proteins $\sim$ MAL.** Top 30 FDR-significant proteins, $\hat{\beta}$ for MAL association with 95% CI.



Supplementary Figure A. Forest plot of the top 30 proteins most significantly associated with MAL score (ranked by FDR). Points show OLS $\hat{\beta}$ (log₂ RFU per unit MAL) with 95% CI, from the model: $\log_2(\text{protein}) \sim \text{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1-PC5}$. Red = positively associated; blue = negatively associated with MAL. Vertical dashed line at zero.

- **Supp Figure B: Forest plot, proteins ~ emphysema progression.** Same 30 proteins, $\hat{\beta}$ for progression association (adjusted for baseline lung density and covariates).



Supplementary Figure B. Forest plot of the same top 30 MAL-associated proteins (ordered as in Supplementary Figure A) showing their association with emphysema progression (Δ CT lung density, HU). Points show OLS $\hat{\beta}$ with 95% CI, adjusted for baseline lung density (`lung_density_vnb_P2`), age, sex, race, pack-years, smoking, scanner, and PC1–PC5. Vertical dashed line at zero.

What Was Not Found / Assumptions Made

1. **FEV₁% predicted:** column `FEV1pp_P2` does not exist in the phenotype file. Not included in Table 1 or any model. Check the data dictionary for the correct column name.
2. **BMI:** no BMI column found. Not included in Table 1 or the clinical regression model. The clinical model adjusts for age, sex, race, smoking, and pack-years only. AIC comparisons may change once BMI is added.

3. **Figure 1 (study schematic):** not automatically generated. Needs to be drawn manually.